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RESEARCH

Hourly Solar Irradiance Forecasting Using Long Short Term Memory and Convolutional Neural Networks

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Abstract
Accurate solar irradiation prediction is directly linked to estimating the power output of photovoltaic systems, making it essential for optimal operation and management of solar facilities. Over the past few years, there has been an increasing interest in the domain of solar irradiance prediction, where numerous long- short-term memory (LSTM) and convolutional neural network (CNN)-based models have emerged as promising approaches for solar irradiance forecasting. This paper aims at developing a simple and accurate LSTM and CNN-based global horizontal irradiance (GHI) forecasting predictor. More specifically, the present study examines the impact of several parameters on the prediction accuracy, including the forecasting timestep length, dataset size, number of LSTM units, number and size of CNN filters, and the input configuration of the forecasting model. Using the Los Angeles solar Irradiance datasets, several LSTM and CNN-based yearly and seasonal models are explored for predicting one-hour ahead solar irradiance. It has been observed through the simulation results that using seasonal models enhances prediction accuracy and that prediction quality increases in stable datasets free of contradictory examples. In CNN-based forecasting models, accuracy is clearly influenced by the prediction timestep length, kernel size, and max pooling, where, after several tests, a prediction lag $N\_steps = 72\ hours$ and a max pooling = 1 have been found to be the optimal parameter values. While stacked LSTM layers improve accuracy, stacking CNN layers does not necessarily lead to improvements. More specifically, the obtained results show that an annual stacked LSTM-based model with three descending layers (32, 16, and 8 units) outperforms all other examined forecasting structures, achieving root mean square error (RMSE) and mean absolute error (MAE) values of approximately 37.08 W/m^2 and 15.23 W/m^2, respectively. However, a seasonal CNN GHI input forecasting model without max pooling might surpass LSTM-based schemes in terms of efficiency.

Keywords Global Horizontal Irradiance (GHI) · Convolutional neural networks (CNN) · Long short-term memory (LSTM)

Introduction

The transition towards renewable energy systems has become a priority for energy security in every country [1]. As a consequence, we observe a significant integration of these new systems into the electricity industry. However, this emerging energy system is complex and requires an in-depth study of resource utilization, energy efficiency, reliability, security, control models, energy management systems, and the economic distribution of electricity. Very comprehensive overviews and research papers have presented a detailed analysis of these themes using advanced intelligence techniques in [2–5]. Among renewable resources, solar energy stands out as a promising source, with production having considerably increased by around 25% compared to 2021 [6]. However, the solar energy market is highly variable and intermittent due to

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fluctuations in solar irradiation, which impacts photovoltaic production. Thus, precise forecasting of solar energy variations is crucial for photovoltaic systems to offer high efficiency and better service. [7].

In the last few years, a significant number of researchers and various companies and institutions have developed different techniques to forecast GHI, which is considered the main basis that directly affects solar PV power production [8, 9]. A variety of solar irradiance forecasting methods and models are utilized, each exhibiting different prediction performances. However, the primary determinant of a specific technique's effectiveness lies in the forecasting horizon and the location under consideration [10]. So, depending on the different approaches used, these methods can be categorized, as illustrated by Fig. 1, into the following three classes:

1. Physical models, including global meteorological and local mesoscale models based on numerical weather prediction (NWP).
2. Image-based methods.
3. Data-driven or time-series models: statistical models (ST), machine learning (ML) models, and deep learning (DL) models.

A fourth class related to hybrid models has been reported in several relevant research works. This kind of approach, considered as a combination of techniques from the same category and/or from different categories, has been introduced to overcome the limitations of each single scheme and improve the overall forecasting accuracy [11–15].

Based on time horizon [16–19], four forecasting types could be distinguished, as well as their applications and their corresponding adapted models, as reported in Table 1.

The physical models, based on NWP, employ physical equations that characterize changes in the atmosphere, where weather observations obtained from ground sensors and satellites are brought into these mathematical models in a daily process using dedicated operational services [20, 21]. These models are provided by the National Weather Service (NWS), which is principally located in Europe and Northern America. In the USA, we find the Global Forecast System (GFS) model from the National Oceanic and Atmospheric Administration (NOAA). The most commonly used models in Central Europe include the European Centre for Medium-Range Weather Forecasts (ECMWF) models, the Deutscher Wetterdienst (DWD) Global Model for Europe (GME) model from Offenbach, Germany, the United Kingdom Meteorological Office (UKMO) model from Bracknell, United Kingdom, and the Limited Area Dynamical Adaptation International Development (ALADIN) model from Toulouse, France. In [22, 23], authors utilized NWP online in real-time to forecast PV power plant output for the following day. These predictions are integrated into optimal control strategies for PV plants, including battery storage systems serving commercial buildings. The main drawbacks of NWP are the initial conditions and the temporal resolution, which is generally more than 1 hour, leading to a coarse model inappropriate for very short-term forecasting [24]. In image-based methods, cloud images, taken by satellites or/and ground-based cameras, are used to predict cloud changes

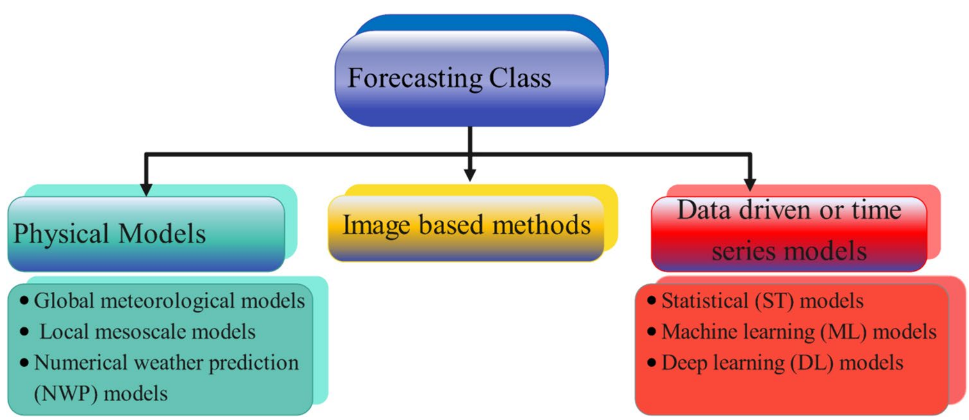


Fig. 1 Forecasting methods classification

Table 1 Forecasting types and applications

Time Horizon	Forecast Type	Application	Adapted model
Few seconds to several minutes ahead	Very short term	Real-time battery storage management	Image methods, ST, ML
1h to 24 h or 1week ahead	Short term	Decision-making, tasks unit commitment	ML, DL, NWP, hybrid models
1 month to 1 year ahead	Medium	Maintenance schedules, power units running	ML, DL, hybrid models
1year to several years	Long term	Strategic power grid operations planning	ML, DL, hybrid models

(position and motion) by applying advanced digital image processing techniques and ML algorithms. These methods, shown to be effective for very short-term prediction horizons, have been applied in various research studies [25–30]. The data driven models use historical measurements to extract some useful information to establish relationships between inputs and the outputs to be predicted. Statistical and ML methods represent the two main approaches of this category, where the size and quality of the historical data play a significant role affecting the forecasting accuracy [18]. Based on mathematical equations, ST models forecast GHI by analyzing the directions in past time series of irradiance as endogenous input features, or, with other variables as exogenous inputs. These variables are considered as a time series that contains three components: long-term trend, periodical components, and the mean. For example but not limited to, ST approaches developed in [31–38] include persistence, Smoothing State Space, Auto-Regressive Integrated Moving Average (ARIMA), Generalized Autoregressive Conditional Heteroskedasticity (GARCH), Seasonal Auto Regressive Integrated Moving Average (SARIMA), regression, and bayesian inference methods. The main drawback of these time series prediction methods is the low accuracy forecasting in short term predictions, moreover, it is essential to test these techniques for stationarity to ensure their proper functionality with non-linear data. According to table 1, ML and DL data-based techniques, can be applied to any time horizon, with no temporal or spatial limitation. In addition, these widely used methods can be applied on line or off line without using any specific physical equipment. In ML approaches, we may find classical learning techniques (SVM, Regression, KNN, Kmeans clustering, etc.), Ensemble learning methods (random forest, XGBoost, etc.) and neural Deep learning networks methods. The provided references [17–19] likely offer comprehensive overviews of ML forecasting methods, including their limitations and accuracy. Despite the fact that deep learning methods are more powerful than typical machine learning techniques and the limitations of a single-model based scheme, we find few publications that use CNN and LSTM networks as autonomous models or in a hybrid structure to predict the GHI [16, 35]. In this introduction, we solely focus on the deep model and review the essence of some works in connection to our considered approach. In [39] an LSTM is applied to predict one hour , one day in advance of GHI. In order to improve prediction accuracy on cloudy days, the clearness-index was introduced as an input data for the LSTM model. ARIMA, SVR, BPNN, CNN and RNN models are compared in varying climate location and complicated weather conditions where the LSTM-based models yield relatively better prediction accuracy. Authors in[40] propose a comparative study between LSTM neural networks, persistence algorithm, linear least square regression, and multilayered feedforward neural networks using backpropagation algorithm (BPNN) for hourly one-day ahead prediction of solar irradiance using data collected on the island of Santiago, Cape Verde. The LSTM technique yields more precise results than the BPNN technique. In [41], the authors predicted Hourly irradiance day+1, for small residential buildings and renewable energy generators by an LSTM neural network. The employed input parameters are day irradiance and day+1 data of temperature, humidity, wind speed, sky cover, and precipitation provided by Korea Meteorological Administration Weather Forecasts. Using data from five regions for learning and a sixth region for prediction, the variation coefficient of RMSE is enhanced by approximately 12% annually compared to other relevant models. Pratima Kumari and Durga Toshniwal [14]investigated the sequential arrangement of the LSTM-CNN scheme as an alternative to the combined CNN-LSTM approach presented in [15]. They utilized identical spatial and temporal feature structures as described in [15]. Furthermore, [14] focuses on 23 sites located in California, while [15] considered 34 locations in Texas. In order to predict the next GHI hour, the authors of [14] demonstrate that the hybrid model, LSTM–CNN, provides a more accurate prediction by testing the forecasting structure for one year, four seasons, and three sky conditions (sunny, cloudy, mixed). In fact, a prediction skill score in the range of 37% to 45% outperforms other hybrid models CNN–LSTM and a few standalone benchmark models (SP, ANN, SVM, LSTM, CNN). Among the hybrid strategies described in the literature are those that perform transformations to time series prior to ML application Xiaoqiao et al. [42] suggested a hybrid model that combined wavelet packet decomposition (WPD), CNN, LSTM, and MLP to forecast solar irradiance one hour in advance. Utilizing 24 hours of recorded metrological data from Denver, Clark, and Folsom, including irradiance, air temperature, relative humidity, and wind speed, different configurations, namely WPD- CNN-LSTM-MLP, WPD-CNN-LSTM, and LSTM are tested. With an RMSE of 32.1 W/m^2, the first model (WPD- CNN-LSTM-MLP) exhibits relatively better performances compared to the other models. According to [43], a Daubechies wavelet transform of order 5 is applied to decompose the solar irradiance time series data (training/validation/test) into the "approximations A3" and "details D1 D2 D3", and the decomposed signals are fed into the Elman Neural Network (ENN) to forecast the new wavelet coefficients in order to reconstruct the solar irradiance data. The proposed model is compared to ENN architecture using Kunming and Denver meteorological data, and two experiment dynamic windows of 0-24h and 8-24h hours irradiance values (1-hour intervals) as inputs, and the 24h+1 as predicted irradiation value. The results reveal that the RMSE in Kunming drops from 68.96 to 26.84 W/m^2 and in Denver from 73.09 to 26.99 W/m^2. J. Z. Bixuan Gao et al [44]

Decomposed the original historical GHI data using the complete ensemble empirical mode decomposition adaptive noise (CEEMDAN) transformation. The generated components are then arranged into five distinct configurations, each of which is applied to CNN-LSTM hybrid model. This study predicts the next GHI hour using data collected from four distinct locations (Los Angeles, Denver, Hawaii's Big Island, and Tamanrasset) over the course of six years and different periods of time. Four years are allocated to training, followed by one year for validation and the remaining year for testing. The best of the five configurations is compared to other related schemes, namely CEEMDAN-LSTM, CEEMDAN-SVM, CEEMDAN-BP, CEEMDAN-ARIMA, and LSTM. The results indicate that the evaluated structure performs better overall in terms of various error indices. Despite the fact that the results of approaches that involve transformation are the best so far in terms of RMSE or MAE, this procedure is not practicable in real applications since future values, required for the reconstruction step, do not exist. This is known as data leakage, and it arises when the causality principle is violated.

As mentioned earlier, many studies have utilized CNN and LSTM deep neural networks or hybrid systems to estimate energy consumption, PV power production, wind speed, and so on. However, there is a notable lack of research applying CNN and LSTM to solar irradiance forecasting. To the best of our knowledge, very few studies have specifically tried to examine the impact of their hyperparameters on prediction quality.

In summary, two key research questions are to be addressed as follows: (1) How to build a simple deep learning models to capture short-term patterns among global horizontal irradiance data; (2) How to effectively identify the best LSTM and CNN hyperparameters and other prediction related parameters to improve the forecasting accuracy.

The scientific contributions of this paper can be summarized as follows:(1) It develops a simple LSTM and CNN-based structures to improve the accuracy and reliability in forecasting global horizontal irradiance; (2) It identifies the optimal LSTM and CNN hyperparameters, input data structure, and the best values for parameters such as the number of LSTM units, the number and size of CNN kernels, dataset size, and prediction window lag to achieve the optimal forecasting scheme; (3) The developed LSTM-based deep learning model enhances GHI forecasting accuracy, enabling operation and maintenance (O&M) managers to make more reliable and efficient decisions.

The remaining sections of the manuscript are structured as follows: Section "Dataset and Preprocessing" provides a brief overview of the dataset outlining and preprocessing. The principles of the basic methods employed in this study are discussed in Section "Methodology". Section "Experiments and Discussion" presents a summary of the obtained simulation results. Finally, Section "Conclusion" concludes the paper.

Dataset and Preprocessing

The meteorological data utilized in this study was obtained from the National Renewable Energy Laboratory (NREL) [https://nsrdb.nrel.gov/]. NSRDB is the National Solar Radiation Database. It is a database fed by real-time meteorological measurements. Indeed, the current learning model is offline, but once the optimal predicting scheme is obtained, it can be deployed for real-time GHI forecasting. For the very first deploying time of the model, we just need to wait for 72 hours (the optimal timestep lag) and the forecasting process can be ensured accurately. The dataset encompasses multiple locations, with the city of Los Angeles in California, situated on the west coast of the United States, being the chosen area of focus.

The choice of Los Angeles was not specifically for the region itself but rather for its representation in the NSRDB database. The NSRDB was selected because it is a comprehensive collection *of hourly and half-hourly* meteorological data and *the three most common measurements* of solar radiation. This data is collected from numerous locations and at various *temporal and spatial scales*, accurately representing regional solar radiation climates. This makes it a rich database that provides valuable insights for *studying solar irradiance.*

The dataset covers a time period from January 1, 1998, to December 31, 2020, with measurements taken at 30-minute intervals. The data includes several variables, such as GHI expressed in units of W/m^2, as well as other meteorological parameters like pressure, temperature, wind speed, relative humidity, and more. Each measurement is accompanied by its respective date and time as illustrated in table 2.

The preprocessing phase involves the following steps:

- Outlier removal: This step focuses on eliminating observations that deviate significantly from the rest of the data. Outliers can negatively impact model estimations, leading to less accurate predictions.
- Handling missing values: In the measurement process, missing values are either removed or replaced with carefully selected values using a benchmark technique. Since time series models necessitate continuous data along the time index, it is not feasible to merely remove observations with missing values and reindex them as if there were no gaps.
- Normalization: The dataset is normalized using the Min-max method, where the normalized GHI value x_N is calculated using its minimum $x_{\min}$ and maximum $x_{\max}$ values as follow:

Table 2 Examples of Los Angeles Solar Irradiance Dataset

Year	Mon	Day	Hour	Min	DHI	DNI	GHI	Clearsky DHI	C_sky DNI	C_sky GHI	Cloud Type	Dew Point	Wind Speed	Humi	Temp	Pressure
1998	1	1	0	0	0	0	0	0	0	0	0	8	1,6	68,17	14	980
1998	1	1	0	30	0	0	0	0	0	0	0	8	1,6	72,72	14	980
1998	1	1	1	0	0	0	0	0	0	0	0	7	1,7	70,92	14	980
…																
2014	11	12	7	0	37	8	38	44	204	66	3	12	2,1	92,21	15	980
2014	11	12	7	30	76	53	86	74	377	150	3	12	2,1	92,21	15	980
2014	11	12	8	0	111	35	121	93	509	241	3	12	2,2	83,15	16	980
…																
2020	12	31	22	30	0	0	0	0	0	0	0	2,9	2,2	65,16	9,1	990
2020	12	31	23	0	0	0	0	0	0	0	0	2	2,4	61,96	8,9	990
2020	12	31	23	30	0	0	0	0	0	0	0	2	2,5	62,81	8,7	990

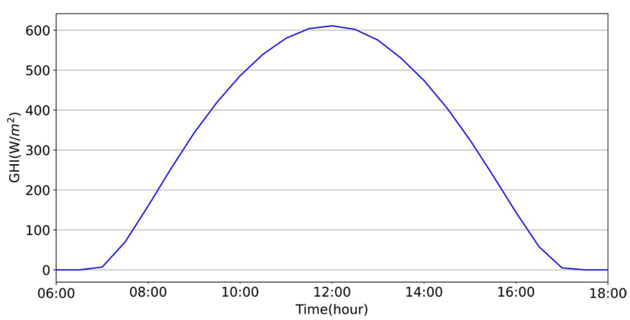


Fig. 2 Distribution of Hourly solar irradiance on June 1st, 2020

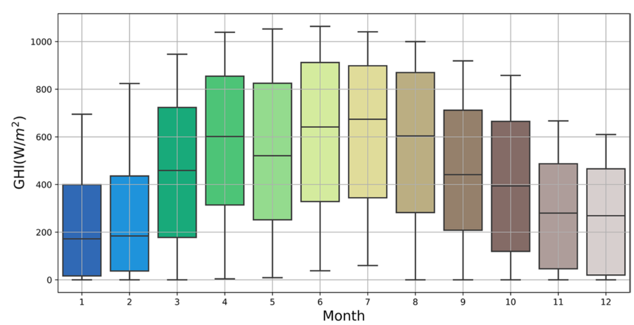


Fig. 3 Monthly solar irradiance distribution of year 2017

$$x_N = \frac{(x - x_{\min})}{(x_{\max} - x_{\min})} \tag{1}$$

Min-max normalization scales all features of the diverse datasets, such as GHI and weather variables like temperature, humidity, and wind speed, to a common range, typically [0, 1]. This uniform scaling ensures consistency across features, which is essential for the efficient convergence of deep learning algorithms, ultimately enhancing model performance. By addressing the non-linearity inherent in the data, min-max normalization allows each input feature to contribute equally to the learning process, thereby reducing bias and improving prediction accuracy. Overall, normalizing the data stabilizes the training process and optimizes the model's predictive capabilities, making it a vital step in building robust DL applications.

Utilizing the preprocessed data as described earlier, Figs. 2, 3, 4 shows randomly selected total amount of daily, monthly, and yearly GHI. Fig. 3 specifically presents a boxplot illustrating the distribution of irradiance for each month in the year 2017. The boxplot reveals a seasonal pattern,

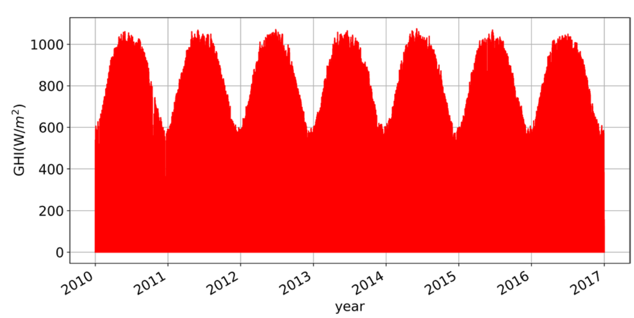


Fig. 4 Yearly solar irradiance distributions

with the median irradiance distribution exhibiting a gradual increase from January to June, followed by a gradual decrease from July to December. Fig. 2 demonstrates a high correlation among the hourly solar irradiance values. The GHI values exhibit a low magnitude at the start of the day, steadily increasing to reach their peak at noon, and then gradually declining during the afternoon

However, when examining the irradiance values across different years, as shown in Fig. 4, no significant differences can be identified.

Methodology

Deep learning neural networks

LSTM

In contrast to a standard neural network that treats all observations in a time series as independent, an LSTM takes into account the sequential dependencies present in the time series. As depicted in Fig. 5, each LSTM cell calculates two outputs at each time step, namely the cell state c_t and the hidden state h_t. This computation is based on the input data x_t, the hidden state from the previous time step h_{t-1} and the previous cell state c_{t-1}. The LSTM update procedure is carried out through three gates: input gate i_t, forget gate f_t, and output gate o_t.

Forget Gate The output f_t of this gate is given by:

$$f_t = \sigma\left(w_f.\left[h_{t-1}, x_t\right] + b_f\right) \tag{2}$$

This gate determines whether information should be retained or discarded. It operates by concatenating the previous state information h_{t-1} with the incoming data x_t using weight matrices w_f and a bias vector b_f. The sigmoid function $\sigma(.)$ is applied to normalize the resulting values between 0 and 1. When the sigmoid output is close to 0, it indicates that the information should be forgotten, while a value close to 1 signifies that the information should be memorized.

Input gate: in this gate a sigmoid σ and a *tan* functions are applied in parallel on the concatenated data $[h_{t-1}, x_t]$ in order to decide which values to write, and which candidate cell $c\prime_t$ to update. The following equations describe the input gate:

$$i_t = \sigma(w_i.\left[h_{t-1}, x_t\right] + b_i) \tag{3}$$

$$c\prime_t = \tanh\left(w_c\left[h_{t-1}, x_t\right] + b_c\right) \tag{4}$$

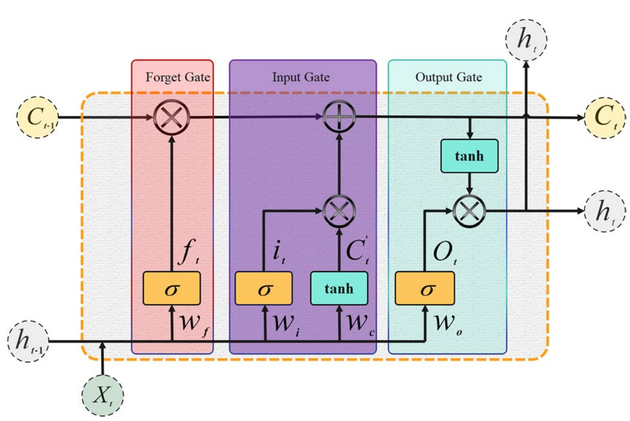


Fig. 5 Structure of LSTM cell

Output gate: governed by equations 4 and 5 the state of the cell can update itself recursively using the forget gate, the input gate, and its old value ($c\prime_t$). The cell state c_t is used to calculate the final cell output h_t.

$$c_t = f_t * c_{t-1} + i_t * c\prime_t \tag{5}$$

$$h_t = o_t * \tanh\left(c_t\right) \tag{6}$$

The predicted output $\hat{y}_t$ is estimated as: $\hat{y}_t = \mathrm{W}h_t$ where W is a weight matrix.

CNN

Convolutional Neural Network is a type of deep learning neural network traditionally used for image classification. This kind of neural network which do not account for sequential dependencies in time series, may use filters to compute correlation between each cell to better understand the relationships among various observations within the time series.

Figure 6 illustrates that a CNN network typically consists of multiple layers that carry out distinct operations on the input data, including convolution, pooling, and activation. The convolutional layer applies a set of kernels to extract features from the input data, while the pooling layer reduces the output size through down sampling. The activation layer introduces non-linearity by applying a non-linear function, such as ReLU. Finally, the fully connected layer at the network end employs the extracted features to make predictions for

the desired output. In our case, the input provided to the network comprises a sequence of time series data

representing global horizontal irradiance. The CNN layers are specifically tailored for 1D data processing, enabling them to handle the sequential nature of the time series data efficiently.

Convolutional Layer For a GHI input sequence $\{x\}$ of length *N*, this layer achieves a convolution operation which can be written for the k_{th} filter as:

$$h_k = f\left(\sum_{i=1}^{N-F+1} W_{k,i} x_{i:i+F-1} + b_k\right) \tag{7}$$

Where F is the length of the filter, $W_{k,i}$ is the weight matrix for the k_{th} filter at position i, $x_{i:i+F-1}$ is the subsequence of $\{x\}$ from position i to position $i+F-1$, b_k is the bias term, and f is the activation function.

Pooling Layer This layer performs a max pooling operation on the output obtained from the convolutional layer, resulting in an output denoted z_k and described by the following equation:

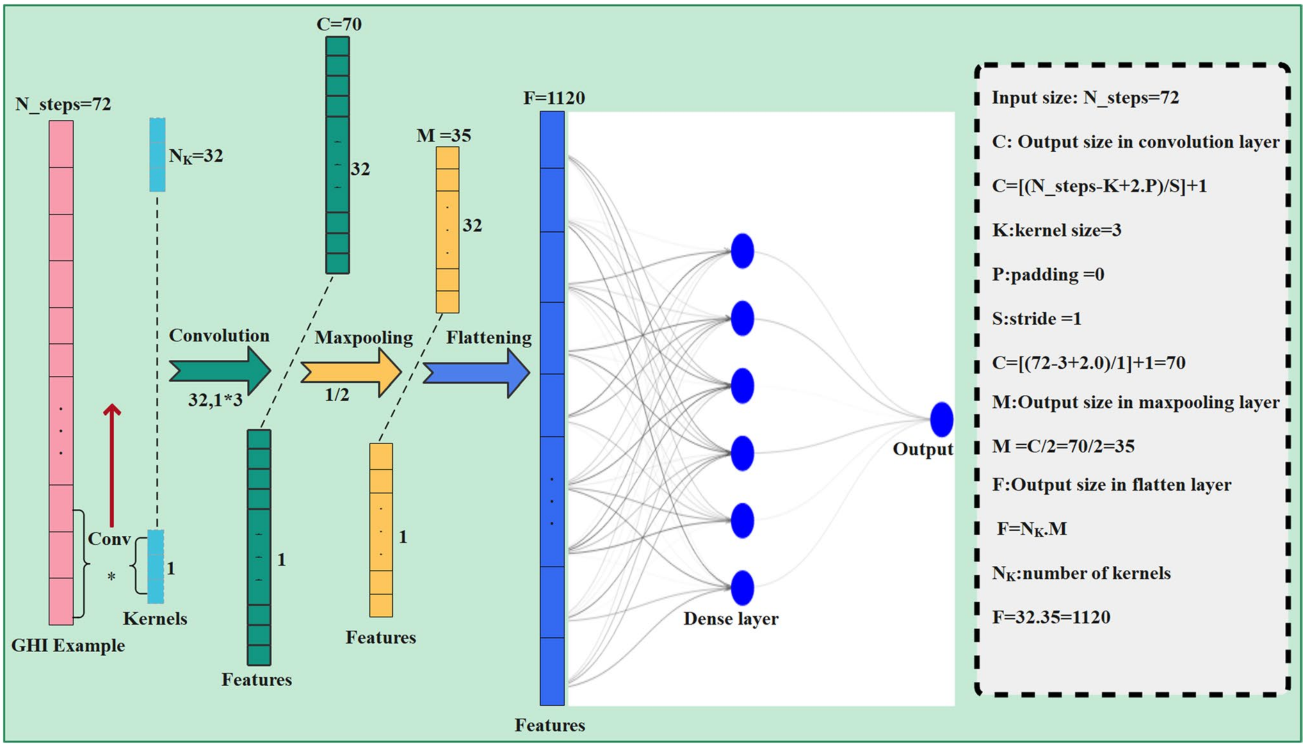


Fig. 6 Structure of a CNN 1D Network

$$z_k = max_{i=1}^{N-F+1} h_{k,i} \tag{8}$$

Fully Connected Layer The output z_k of the pooling layer is flattened and passed through the fully connected layer which applied an activation function g on the sum $(z + b)$, with W is weight matrix, and b is the bias term. The equation is given by:

$$y = \mathrm{g}(Wz + b) \tag{9}$$

During training, the weights of each layer are adjusted in each iteration using back propagation and gradient descent to minimize the error between the predicted GHI and the true GHI. The process is repeated until the CNN converges to a set of weights that minimize the loss.

Forecasting approaches

In this study, two approaches are used to predict the GHI. The first approach uses only historical GHI records, while the second one combines the use of historical GHI data with meteorological parameters. Thus, the described LSTM and CNN models are used to forecast the next value of hourly GHI using fixed timestep. Fixed timestep offers several advantages over sliding windows, including simplicity, consistency, and uniform interval for structuring time series data, making it easier to manage and understand.

The first model predict the next hour x_{t+1} based on *N_steps* passed hours as explicated by this equation

$$x_{t+1} = f\left(x_{t-N_{\_steps}}, x_{t-N_{\_steps}+1}, \cdots x_t\right) \tag{10}$$

Where:

$x_{t+}1$	is the forecasted value of GHI (next hour)
$x_{t\text{-}N\_steps}$	is the previous value of GHI
x_t	is the actual GHI
N_steps	The size of the sequence the timestep
f	is a functional dependency between past and future samples assured by the DL model
t	is the actual hour

An example of inputs and output samples is shown in table 3. Input covers hours 1 to 7 the target is the 8th hour. After making a prediction, the sliding window shifts forward

Table 3 Example of GHI input and output samples

Input	Sample1	Sample2	Sample3	Sample4	Sample5	Sample6
	0	0	0	0	0	0.02581
	0	0	0	0	0.02581	0.189582
	0	0	0	0.02581	0.189582	0.393712
	0	0	0.02581	0.189582	0.393712	0.589395
	0	0.02581	0.189582	0.393712	0.589395	0.755045
	0.02581	0.189582	0.393712	0.589395	0.755045	0.879399
Output	0.189582	0.189582	0.393712	0.589395	0.755045	0.879399

by 1 hour. The new window covers hours 2 to 8 and the target become hour 9. This approach allows the model to continuously learn from the most recent data while making predictions for the next time step. It is effective in capturing temporal dependencies and trends in time series data.

Also, six input features namely, GHI, temperature, humidity, wind speed, pressure, and clearness index are used in all experiments. These features were selected using the wrapper method, involving an exhaustive exploration of different feature combinations to determine the best fit for the specific model. This approach is described as follows:

$$x_{t+1} = f([x_t], [x_{1t}], [x_{2t}], [x_{3t}], [x_{4t}], [x_{5t}]) \tag{11}$$

Where:

x_{t+1} is the forecasted value of GHI

$[x_t]$ is vector of *N_steps* passed hours GHI

$[x_{1t}]$ vector of *N_steps* passed hours Temperatures

$[x_{2t}]$ vector of *N_steps* passed hours Wind speed

$[x_{3t}]$ vector of *N_steps* passed hours Pressures

$[x_{4t}]$ vector of *N_steps* passed hours Humidity

$[x_{5t}]$ vector of *N_steps* passed hours Clearness index

In this study, we conducted several experiments, each with its own objective and dataset. The various datasets are divided into training, validation, and testing sets using the following steps:

- Training Data: For each experiment, the training data is defined over a specific period. For example, from 2010 to 2016.
- Validation Data: The year following this training period is used as the validation data. In this example, the validation data corresponds to the year 2017.
- Testing Data: The year following the validation period is considered the test data. In our case, this is the year 2018.

In all experiments, the data is generally distributed as 75% for training, 12.5% for validation, and 12.5% for testing.

Figure 7 shows the flowchart of the forecasting method based on DL models:

According to the above flowchart, the forecasting model development steps are carried out as follows:

1. Read data.
2. Preprocess data: handle the missing data, remove the outliers and perform normalization.
3. Select the most relevant features for consistency.
4. Choose the dataset to use and split it into training, validation, and unseen testing sets.
5. Build the appropriate LSTM or CNN-based forecasting architecture.
6. Train and validate the model.
7. Optimize hyperparameters to improve model performance.
8. Repeat steps 5 and 6 until the desired model is obtained or stopping criteria are met.
9. Assess and validate the obtained model using unseen testing data.
10. Choose the final model based on test metrics that reflect its generalization capability.

Performance Metrics

To evaluate the accuracy of forecasting, several metrics are used, namely the MAE (W/m^2), RMSE (W/m^2), Mean Absolute Percentage Error MAPE (%), and Pearson Correlation Coefficient (r) whose expressions are, respectively, given by Equations 12:15 where x_i and $\hat{x}_i$ denote the measured and estimated GHI values, respectively

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|x_i - \hat{x}_i\right| \tag{12}$$

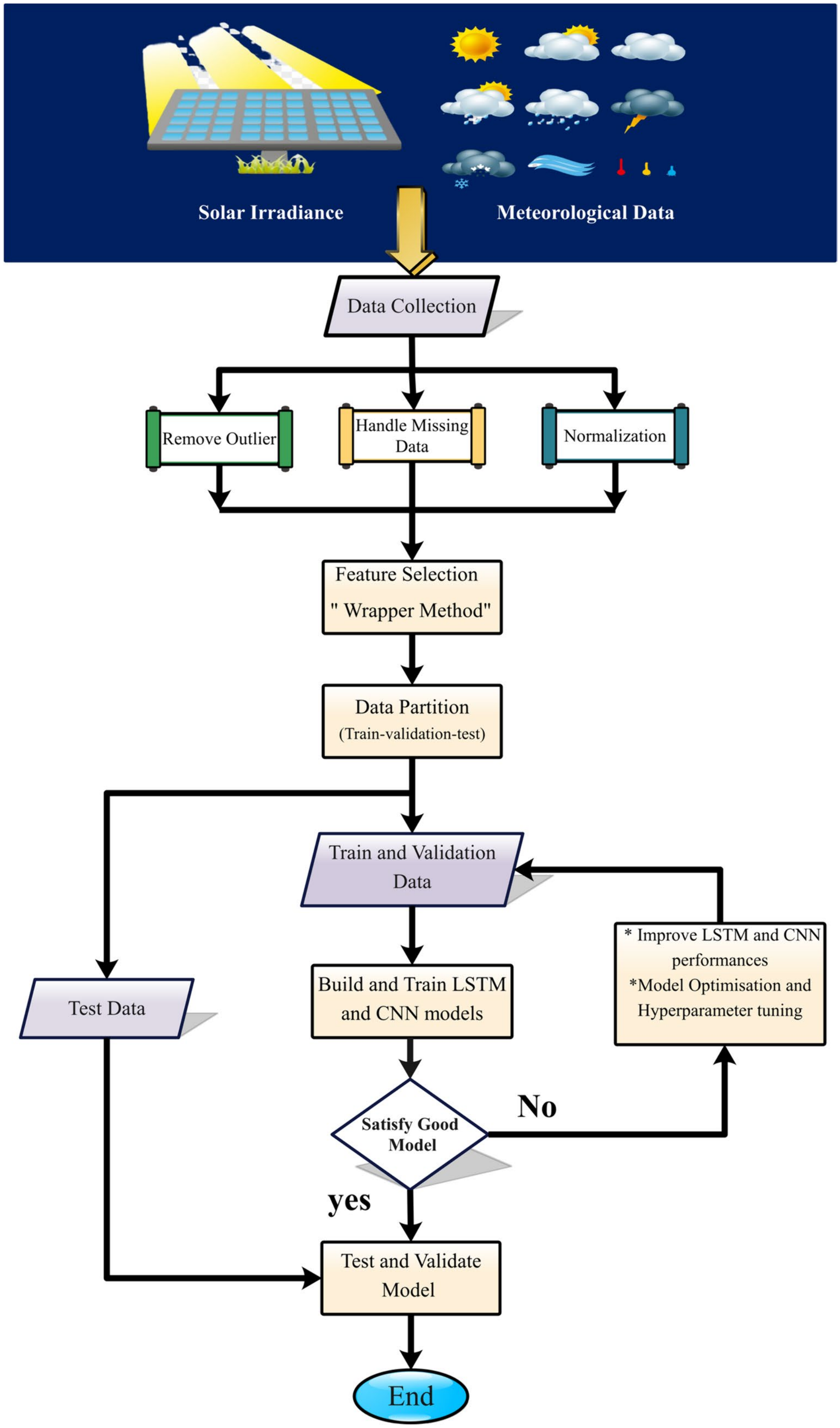


Fig. 7 Flowchart of the forecasting method

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(x_i - \hat{x}_i\right)^2} \tag{13}$$

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{x_i - \hat{x}_i}{x_i}\right| \tag{14}$$

$$r = \frac{\sum_{i=1}^{N}(x_i - \overline{x})(\hat{x}_i - \overline{\hat{x}})}{\sqrt{\sum_{i=1}^{N}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{N}(\hat{x}_i - \overline{\hat{x}})^2}} \tag{15}$$

Experiments and Discussion

This paper conducts multiple simulation experiments to analyze how different parameters impact the accuracy of predictions. It is important to highlight that the parameters are considered in all the tests:

- The training data is sampled every 30 minutes. The objective is to predict the average hourly GHI value.
- The prediction focuses on solar irradiance one hour in advance.
- The optimization of the hyperparameters for both LSTM and CNN-based deep learning networks (such as learning rate, dropout rate, batch Size, activation function, and the momentum) is accomplished through the use of the Random Search Keras Tuner approach and employing the Early stopping callback with *monitor = "validation loss''* and a patience equals to 25.
- For the other hyperparameters, including the number of LSTM units, the CNN number of filters and kernel size, the grid search technique, provided in the scikit-learn library, has been adopted with 5-fold cross validation.
- To accomplish this, the following setting values have been adopted:
  - Epochs: 1000
  - Optimizer: Adam
  - Loss function: Mean Squared Error (MSE)
  - Metrics: Mean Absolute Error (MAE)

Annual Performance

First, the paper examines the impact of timestep, database, LSTM, and CNN parameters on the annual performance. Subsequently, the optimal configuration of the two deep networks is evaluated across four different seasons: spring, summer, autumn, and winter.

Effect of Timestep and Number of LSTM Units

In this test, the performance of the prediction is examined in relation to two factors: the length of timestep (N_steps) and the number of LSTM units (N_units). The training dataset is composed of GHI values from 01/01/1998 to 31/12/2018. The validation dataset comprises the data from 01/01/2019 to 30/12/2019, while the test dataset consists of the data from the year 2020.

Table 4 and Figs. 8, 9 illustrate the impact of varying the number of LSTM units from 16 to 128, while keeping N_steps fixed. The RMSE and MAE show slight changes across the range of N_units. For instance, when $N\_units \in [16 - 128]$ and $N\_steps = 7h$, the maximum RMSE is $44W/m^2$ and the minimum RMSE is $43.88W/m^2$.

Conversely, experimenting with different N_steps values within the range of [7h to 72h] while maintaining a fixed number of LSTM units leads to significant improvements. For example, when $N\_steps = 12$ and $N\_units = 16$, the RMSE is 44.29 W/m^2, whereas with $N\_steps = 72$ and $N\_units = 32$, the RMSE drops to 40.94 W/m^2. Thus, it can be concluded that the hyperparameters "number of units" in the LSTM has a negligible influence on the forecasting quality due, possibly, to the overfitting or diminishing returns model capacity when varying LSTM units. On the other hand, selecting an appropriate timestep length for the GHI series can greatly enhance the prediction quality.

Table 4 Effect of timestep and LSTM units

		RMSE (W/m^2)				MAE (W/m^2)			
Number of units units		16	32	64	128	16	32	64	128
N_steps	7	44,13	43,96	43,81	43,88	18,06	17,86	19,09	16,84
	12	44,29	43,55	43,3	43,16	17,9	16,71	16,75	16,84
	24	42,41	41,47	42,04	41,6	15,23	14,47	16,06	16,31
	36	41,2	41,4	41,84	41,58	14,9	16,89	16,24	15,09
	48	41,17	41,16	41,19	41,32	14,69	15,23	15,93	17,07
	60	41,82	41,35	41,81	41,18	16,88	15,31	16,7	15,46
	72	41,21	40,94	41,33	41,34	14,99	14,23	14,72	14,93

Remarkably, using a small window size can result in a scarcity of historical information, which, in turn, complicates the task of making precise forecasts for the upcoming hour GHI. However The use of a longer sliding window (N_steps = 72 hours) improves forecast accuracy, probably, by providing more contextual information, smoothing fluctuations, improving feature representation, capturing complex temporal relationships and guarantees statistical robustness.

Impact of Training Dataset Size

The previously achieved optimal configuration ($N_{steps} = 72h$, $N\_units = 32$) has been utilized in two types of periods: sequentially ordered periods ranging from 2015-2018 to 1998-2018, and randomly selected periods, as reported in Table 5.

The results reported in table 4 and Figs. 10, 11 provide clear evidence that the model accuracy remains consistent regardless of the length of the input time series.

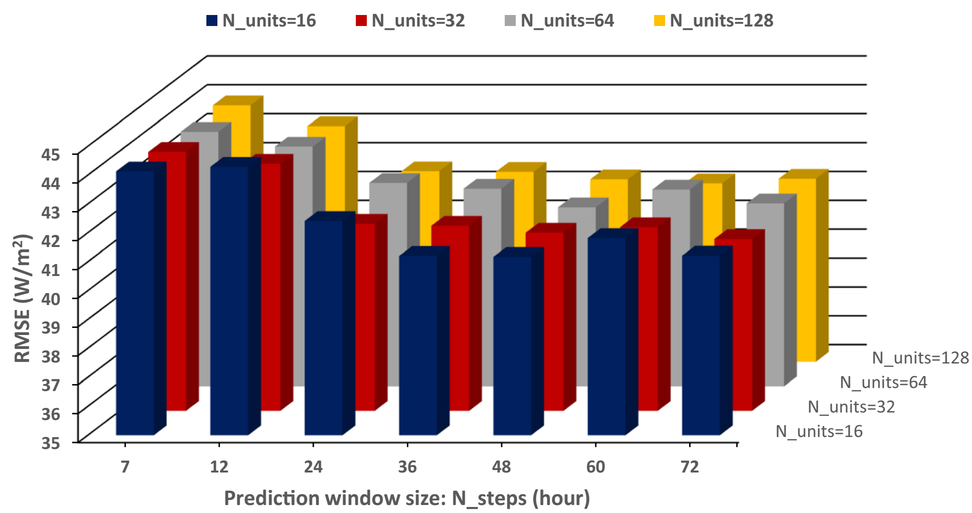


Fig. 8 RMSE in terms of timestep length and number of LSTM units

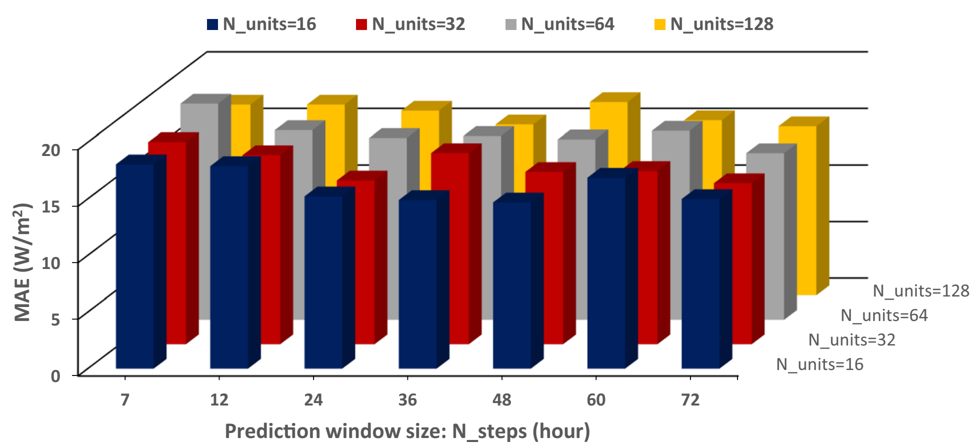


Fig. 9 MAE in terms of timestep length and number of LSTM units

Table 5 Effect of training dataset length

Ordered periods	RMSE (W/m^2)	MAE (W/m^2)	Random periods	RMSE (W/m^2)	MAE (W/m^2)
01/01/2015-31/12/2018	40.83	16.11	05/05/2010-04/05/2016	**38.71**	16.73
01/01/2011-01/01/2018	41.05	15.34	01/01/2012-31/12/2015	42.6	16.58
01/01/2008-31/12/2018	41.06	15.35	01/01/2008-31/12/2012	**39.92**	16.62
01/01/2004-31/12/2018	41.08	15.86	01/01/2004-31/12/2009	47.46	19.7
01/01/2000-31/12/2018	41.52	15.83	01/01/2015-31/12/2018	40.83	16.11
01/01/1998-31/12/2018	40.94	14.23	05/05/2008-04/05/2014	49.53	20.24

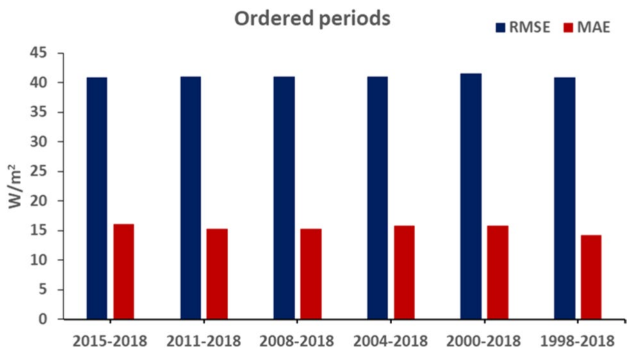


Fig. 10 Effect of training dataset length for Ordered periods

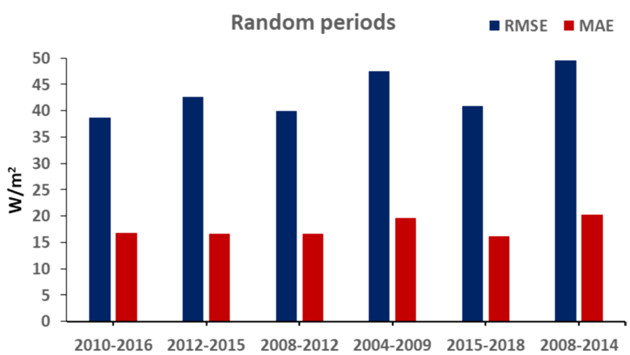


Fig. 11 Effect of training dataset length for random periods

Specifically, the RMSE values of 38.71 W/m^2 and 39.92 W/m^2 observed during the separate periods of 2010-2016 and 2008-2012, respectively, are lower than the RMSE value for the period 1998-2018.

One possible explanation for this discrepancy is that the quality of predictions is not solely determined by dataset length; rather, it is significantly influenced by the quality and relevance of the data. While longer time series data can enhance learning, they may also introduce fluctuations and contradictory examples *(where the same entries have different labels),* that compromise prediction accuracy by introducing additional confusion during the training process. For instance, our study shows that the RMSE for the dataset from 1998 to 2000 is 44.45 W/m^2, compared to 38.45 W/m^2 for the dataset from 2010 to 2016, suggesting that a shorter dataset can yield higher precision when it captures essential trends without excessive noise. Ultimately, a well-selected shorter dataset may outperform a longer one filled with inconsistencies. Therefore, when implementing a forecasting model, it is crucial to assess data quality and consistency by analyzing for fluctuations and contradictions to determine the most suitable dataset length for accurate predictions.

Influence of Input Parameters Form and Stacked LSTM Layers

In this simulation, the same data configuration used in the previous experiment is maintained, which consists of training data collected from 05/05/2010 to 04/05/2016. The parameters used for this configuration include *N_steps* = 72*h* and LSTM units = 32, resulting in the best-achieved results with an RMSE value of 38.71 W/m^2. The objective is to analyze the impact of input parameters on the prediction, and for this purpose, three different variants are considered:

- In the first variant, the input sequence is represented as a one-dimensional vector containing solely the measured GHI values. These values are organized in units of 72 hours, giving an input shape of (72,1).
- The second variant involves representing the input as a matrix with a shape of (*N_steps, N_features*), where *N_steps = 72h* and *N_features = 6*. The six features of meteorological weather, including GHI, temperature, humidity, wind speed, pressure, and clearness index, are considered in this matrix representation. Therefore, the input shape becomes (72, 6).
- In the third variant, the input data is transformed into a one-dimensional vector with a shape of *(N_steps×N_features, 1) = (72×6, 1),* combining all the input features from variant 2 into a single vector.

According to the results presented in Table 6, incorporating all meteorological parameters during training leads to a slight enhancement in the LSTM model's performance compared to training with only GHI. The resulting RMSE and MAE are reduced to 38.08 W/m^2 and 16.45 W/m^2, respectively. The LSTM is able to effectively integrate the temporal GHI patterns with the spatial relationships between GHI and meteorological variables. Additionally, employing a stacked architecture with three descending layers (32/16/8) contributes to further improvements in forecasting performance. It is possible that this decreasing architecture (32, 16, 8) enables the LSTM to learn more complex representations, reduce overfitting, enhance gradient flow, and capture temporal dependencies effectively. As a result, the RMSE is reduced to approximately 37.08 W/m^2, and the MAE decreases to around 15.23 W/m^2.

Effect of CNN Hyperparameters

To examine the effect of different structural parameters on prediction accuracy, multiple simulation experiments were conducted using a one-dimensional convolutional neural network (1D CNN) architecture. Using 100 iterations, the aim was to investigate the impact of various CNN structure parameters, including the number of filters, kernel size, and pooling, on the accuracy of predictions. The input sequence for these experiments comprised GHI values, with a timestep of 72 hours. This timestep was determined to be the optimal lag based on previous experiments. The experiments were carried out for three different values of max pooling (1, 2, and 3) and five different values of filters (16, 32, 64, 128, and 256). Notably, a max pooling value of 1 indicated that no max pooling was applied. Additionally, various kernel sizes (1, 2, 3, 4, 5, and 6) were combined with the filters to assess their impact on the prediction accuracy.

The obtained results reported in Tables 7 and 8, as well as Figs. 12, 13, 14 supported by statistical calculations, demonstrate the influence of different parameters on the RMSE and MAE for the prediction model. Specifically, in terms of the mean and standard deviation, it can be concluded that the best and most stable results for RMSE and MAE were obtained when max pooling was set to 1. The mean RMSE for this case was approximately 39.03 W/m^2, with a standard deviation of 0.31 W/m^2, indicating consistent and accurate predictions. In contrast, for max pooling values of 2 and 3, the mean RMSE values were higher (44.06 W/m^2 and 50.87 W/m^2, respectively) with significant deviations (standard deviations of 12.06 and 7.66, respectively).

The observed stability with max pooling = 1 is interesting and relevant. This stability is likely attributed to the sampling effect, where the shape of the output filters is preserved when transferred to the fully connected layer. Max pooling reduces the spatial size of feature maps by retaining only the maximum value in each window. With max pooling = 1, this size reduction does not occur, as each window contains only one value. Thus, the shape of the feature maps is effectively preserved when transitioning to the fully connected layer.

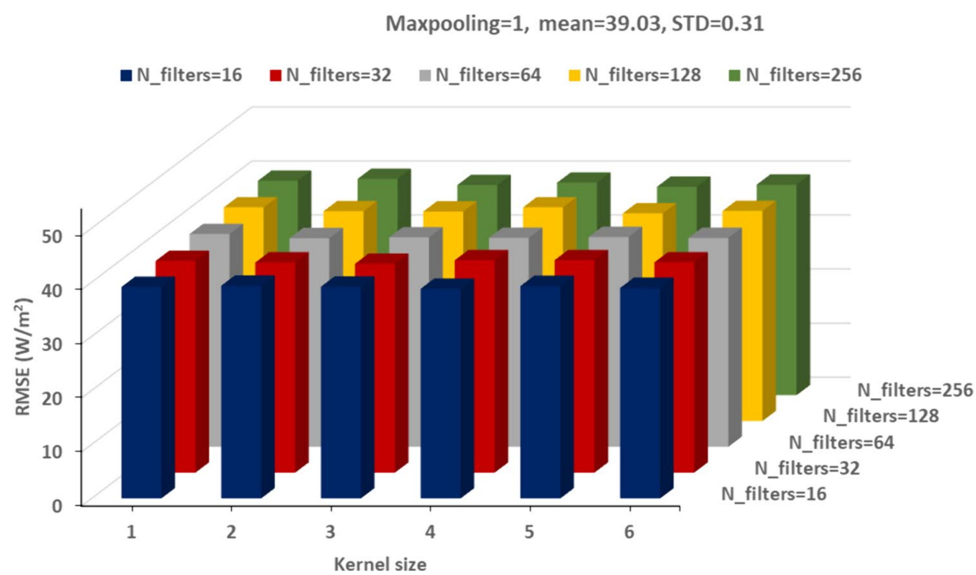


Fig. 12 RMSE versus number and size of filters for max pooling=1

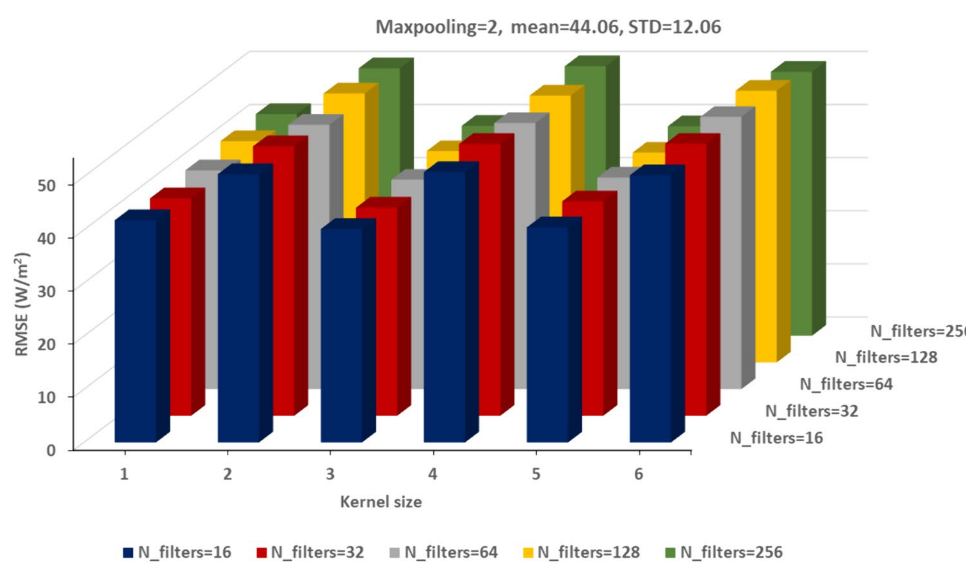


Fig. 13 RMSE versus number and size of filters for max pooling=2

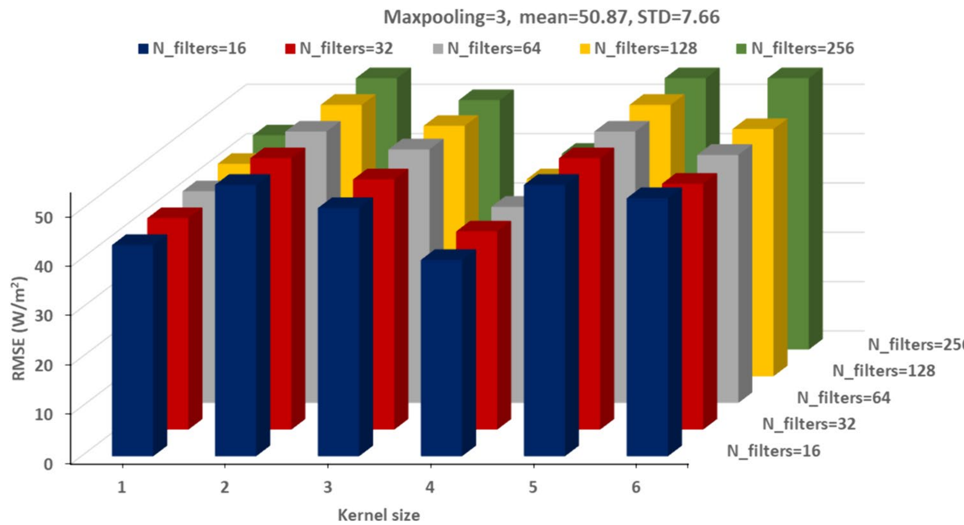


Fig. 14 RMSE versus number and size of filters for max pooling=3

Table 6 Effect of input sequences and LSTM Layers

Input Sequence	Units per layer	RMSE (W/m^2)	MAE (W/m^2)
GHI	32	39.19	16.87
Matrix (72,6)	32	38.08	16.45
Vector (72×6)	32	44.6	25.04
GHI	32/64	38.9	17.07
GHI	32/16	38.46	16.56
GHI	32/16/8	38.12	15.17
Matrix (72,6)	32/16	37.72	15.54
Matrix (72,6)	32/16/8	37.08	15.23
Vector (72×6)	32/16	40.61	18.54
Vector (72×6)	32/16/8	42.39	20.56

However, for higher max pooling values, in particular max poling = 2, the alternation of good and bad performance according to kernel size parity is very marked, with odd sizes giving MAE and RMSE significantly lower than even size values.

Additionally, Tables 7 and 8 reveal that when the number of filters is varied from 16 to 256 for a fixed kernel size ranging from 1 to 6, the RMSE remains unchanged, accompanied by low standard deviation. This suggests that the number of filters does not significantly impact the accuracy of the predictions. However, when the filter size is varied from 1 to 6 for a fixed number of filters selected from 16 to 256, the RMSE values change, and the standard deviation increases. This highlights that the size of filters does affect the forecasting precision. Indeed kernel size can significantly affect the model's ability to capture features. Larger kernels may capture broader patterns, while smaller ones can focus on finer details, providing a more global view of the input data. The general conclusions indicate that max pooling and kernel size are the most influential hyperparameters, with max pooling =1 yielding the best results. This suggests that for the task under consideration, it is preferable to use a max pooling of 1 and an odd kernel size, with the number of filters being of less importance. The optimal configuration seems to be max pooling of 1 with 64 or 128 filters and an odd kernel size equals to 3 or 5.

Influence of Input Parameters Form and Stacked CNN Layers

Table 9 shows the RMSE and MAE errors for the stacked CNN layers obtained by employing various input configurations previously utilized in Section "Influence of input parameters form and stacked LSTM layers". These configurations include max pooling = 1, filter size = 3, and number of filters = 32. The results reveal that the model with a single layer and GHI as input yields the lowest RMSE and MAE values (38.41 W/m2) among all the models. It may be possible to predict GHI more accurately by utilizing GHI observations alone as input data. The autocorrelation and cyclical character of GHI may be captured by the DL models without the requirement for extra inputs from meteorological data.They can successfully deduce the dynamics and temporal patterns of GHI just from past measurements.

For a more comprehensive illustration of the annual model efficacy, Figs. 15 and 16 present a detailed analysis of one-day and one-week forecasting for the year 2018 using both LSTM and CNN models. The depicted curves of forecasted and actual GHI data demonstrate a remarkable degree of resemblance throughout the entire prediction period, providing clear evidence of the accuracy and performance of the forecasting models.

Table 7 RMSE and statistics for different hyperparameters of CNN

Kernel size		Maxpooling=1						Maxpooling=2						Maxpooling=3					
		1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6
N° of filters	16	39.13	39.35	39.18	38.79	39.29	38.94	41.82	50.48	40.15	50.86	40.47	50.28	42.82	59.30	50.31	39.79	59.22	52.30
	32	39.16	38.88	38.64	39.30	39.30	38.89	40.91	50.71	39.16	51.14	40.33	51.17	42.86	60.57	50.72	40.16	59.06	49.84
	64	39.38	38.59	38.74	38.65	38.82	38.64	41.15	49.72	39.39	50.07	39.82	51.22	42.88	59.90	51.37	39.75	60.21	50.26
	128	39.53	38.82	38.76	39.53	38.41	38.87	41.64	50.6	39.75	50.21	39.48	51.12	43.05	59.62	50.75	39.92	58.95	50.10
	256	39.68	40.04	38.90	39.33	38.56	38.96	41.71	50.33	39.51	50.72	39.41	49.66	43.42	60.48	50.56	39.69	59.13	59.13
MEAN	**39.03**							**44.06**						**50.87**					
STD	**0.31**							**12.06**						**7.66**					

Table 8 AME and statistics for different hyperparameters of CNN

Kernel size		Maxpooling=1						Maxpooling=2						Maxpooling=3					
		1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6
N° of filters	16	16.12	17.45	16.47	15.66	15.99	15.53	17.35	21.86	17.78	21.46	17.18	20.63	17.93	25.89	20.98	16.30	24.63	22.96
	32	16.12	16.41	15.26	15.61	15.77	15.52	16.28	21.62	16.08	22.59	17.08	21.23	18.20	26.42	23.66	18.07	25.22	20.81
	64	16.07	15.59	15.62	15.45	15.55	15.54	16.67	19.96	15.48	20.34	16.97	22.87	17.91	25.30	21.74	15.63	26.67	21.71
	128	16.64	16.32	15.48	16.32	15.21	16.84	17.52	20.72	16.70	20.66	16.07	21.72	17.20	26.40	21.70	16.29	25.41	20.67
	256	16.91	18.56	15.55	16.71	16.3	16.2	16.98	21.04	16.31	21.06	15.42	20.25	17.60	26.08	20.30	16.24	26.08	26.83
MEAN	**16.09**							**18.82**						**214.69**					
STD	**0.72**							**2.43**						**3.82**					

Table 9 Effect of input sequences and CNN layers

Input Sequence	CNN layers	RMSE (W/m^2)	MAE (W/m^2)
GHI	CNN1	38,41	15,.21
Matrix (72,6)	CNN1	43,69	19,77
Vector (72×6)	CNN1	42,009	17,79
GHI	CNN1-CNN2	38,68	18,08
GHI	CNN1-CNN2-CNN3	38,99	16,76

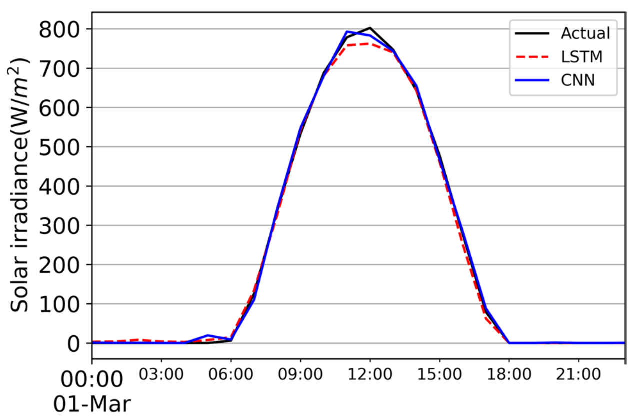


Fig. 15 Actual and predicted GHI for one day forecast

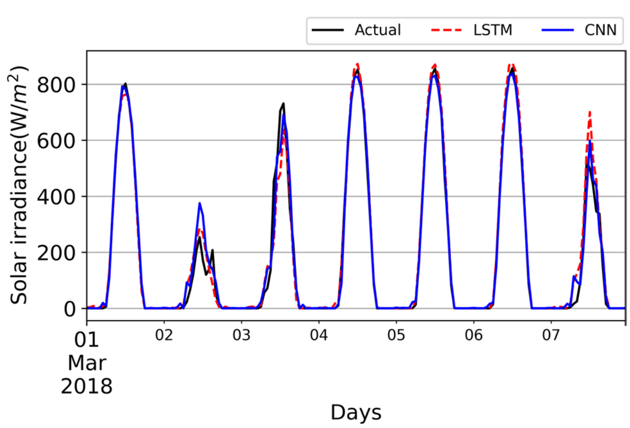


Fig. 16 Actual and predicted GHI for one week forecast

Seasonal Performance

This section investigates the seasonal prediction of GHI using the optimal configurations obtained from previous experiments, including both CNN and LSTM. The simulation involves N_steps = 72 hours, stacked LSTM layers (32/16/8) with an input matrix comprising six meteorological features, as well as a 1D single CNN layer using a sequence of GHI values as input. Two datasets are employed: one covering the period from 2010 to 2016, and the other spanning from 1998 to 2018. The obtained RMSE, MAE, MAPE, and *r* of the CNN and LSTM models for the four seasons (spring, summer, autumn, and winter) are presented in Table 10 and 11 accompanied by Figs. 17 and 18.

Upon analysis, it can be easily observed that in winter, the RMSE for LSTM In the 2010–2016 dataset, it is 25.85 W/m^2, while for CNN, it is slightly higher at 26.75 W/m^2. The MAE values are 11.85 W/m^2 for LSTM and 11.23 W/m^2 for CNN, indicating that CNN performs slightly better in terms of MAE. In the 1998–2018 data set, both models show a deterioration in RMSE, with LSTM at 28.12 W/m^2 and CNN at 27.51 W/m^2, but CNN maintains a lower MAE (10.40 W/m^2) compared with LSTM (11.86 W/m^2). In spring, the results show a notable increase in prediction errors for both models. In the 2010–2016 dataset, LSTM has an RMSE of 48.27 W/m^2 and an MAE of 25.15 W/m^2, while CNN's RMSE is 49.96 W/m^2 and its MAE is 24.00 W/m^2. In the 1998-2018 dataset, LSTM's RMSE increases to 53.17 W/m^2 with a MAE of 27.36 W/m^2, while CNN shows a slight improvement in RMSE (48.70 W/m^2) but a higher MAE (16.93 W/m^2). This suggests that both models struggle to make predictions in spring, but that LSTM shows a greater increase in error. In the autumn, both models perform relatively better than in the spring. In the 2010–2016 dataset, LSTM's RMSE is 37.31 W/m^2 and MAE is 15.82 W/m^2, while CNN performs better with an RMSE of 36.89 W/m^2 and MAE of 13.29 W/m^2. In the 1998-2018 dataset, both models have similar RMSE values (28.11 W/m^2 for LSTM and 28.22 W/m^2 for CNN), but CNN has a lower MAE (10.26 W/m^2) compared to LSTM (10.52 W/m^2), indicating a consistent advantage for CNN in autumn. In summer, the models show a noticeable increase in RMSE values. For the 2010–2016 dataset, LSTM has an RMSE of 43.30 W/m^2 and a MAE of 19.68 W/m^2, while CNN shows slightly better performance with an RMSE of 42.22 W/m^2 and a MAE of 17.86 W/m^2. In the 1998–2018 dataset, LSTM's RMSE reaches 48.17 W/m^2 with a MAE of 18.86 W/m^2, while CNN's RMSE is 47.50 W/m^2 with a MAE of 16.71 W/m^2. This trend indicates that, although both models struggle in summer, CNN still outperforms LSTM in terms of error. The analysis clearly shows that CNN generally outperforms LSTM during most seasons, particularly in terms of MAE, which is a crucial measure for assessing prediction accuracy. The results suggest that while both models show performance variability with seasonal changes, the ability of the CNN to maintain lower error rates can be attributed to its architecture, which effectively captures the spatial characteristics of the data.

The overall forecast errors are quite similar for both models and datasets. The RMSE and MAE errors are lower for winter and autumn, while they are higher for spring and summer. This disparity may likely be due to fluctuations in seasonal cloudiness cycles during spring and summer. DL

Table 10 LSTM and CNN seasonal performance (RMSE, MAE)

Training dataset	Season	LSTM		CNN	
		RMSE (W/m2)	MAE (W/m2)	RMSE (W/m2)	MAE (W/m2)
2010-2016	Winter	25,85	11,85	26,75	11,23
	Spring	48,27	25,15	49,96	24
	Autumn	37,31	15,82	36,89	13,29
	Summer	43,3	19,68	42,22	17,86
1998-2018	Winter	28,12	11,86	27,51	10,40
	Spring	53,17	27,36	48,7	16,93
	Autumn	28,11	10,52	28,22	10,26
	Summer	48,17	18,86	47,50	16,71

Table 11 LSTM and CNN seasonal performance (MAPE, *r*)

Training dataset	Season	LSTM	
		MAPE (%)	*r*
2010-2016	Winter	2.46	0.99
	Spring	11.90	0.98
	Autumn	1.79	0.99
	Summer	3.70	0.99
1998-2018	Winter	3.31	0.99
	Spring	5.18	0.99
	Autumn	2.22	0.99
	Summer	2.48	0.99

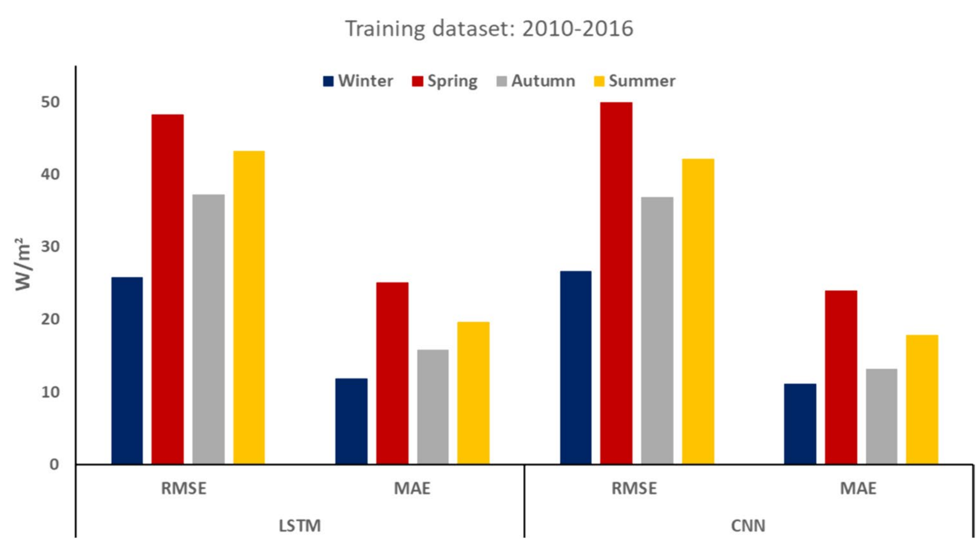
outcomes are heavily reliant on training datasets, and predictions are frequently difficult during these periods.

To enhance the clarity of the results, the model based on the LSTM is evaluated using two additional metrics: Mean Absolute Percentage Error MAPE and Pearson Correlation Coefficient r . The results are presented in Table 11. It is observed that the model's accuracy aligns closely with the trends indicated by the two previously discussed metrics, RMSE and MAE (Table 10) across both datasets. Notably, there is an almost perfect correlation, with a coefficient of r=0.99.

Figures 19 and 20 offer a thorough depiction of the effectiveness of the seasonal model, specifically highlighting one-day and one-week forecasting for the year 2020. It is readily apparent that both the CNN and LSTM models closely track the variations in actual GHI data.

The simulation experiments are conducted in Python on a PC with an Intel Core i5 1.19-GHz central processing unit and 8GB of RAM. Tables 12 and 13 show the summaries and the trainable parameters of the two best models in this study. Also, the total parameters for the LSTM and the CNN-based forecasting models and training time are reported in Table 14. Generally, the number of parameters in a model influences the training time because the optimization algorithm needs to update these parameters during the training process. As the number of parameters increases, the computational complexity of the optimization algorithm also increases, resulting in longer training times. Even though the LSTM-based structure has the fewest parameters, it can be seen that the LSTM-based scheme

Fig. 17 Seasonal forecasting performance for 2010-2016 dataset



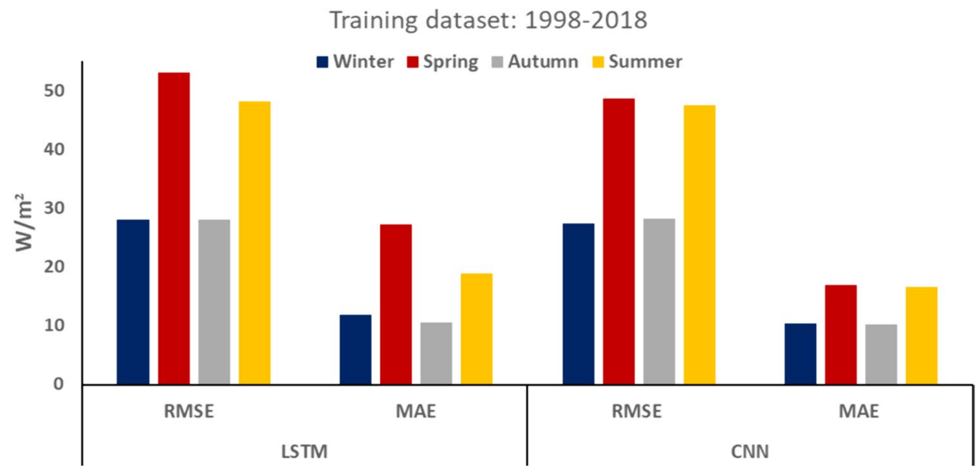


Fig. 18 Seasonal forecasting performance for 1998-2018data-set

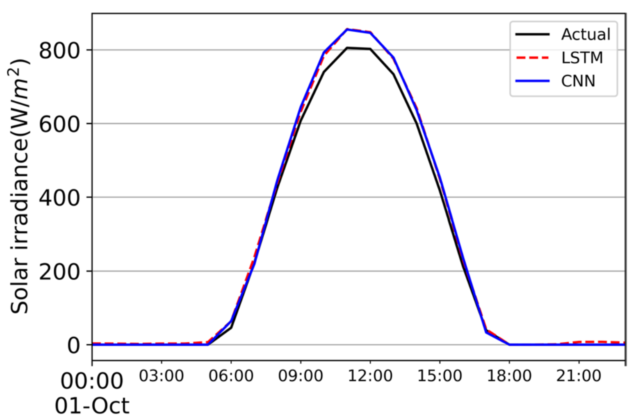


Fig. 19 One day GHI forecasting using seasonal model

Table 12 LSTM-based model summary

Model: "sequentiL"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 72, 32)	4992
lstm_1 (LSTM)	(None, 72, 16)	3136
lstm_2 (LSTM)	(None, 72, 8)	800
Flatten (Flatten)	(None, 576)	0
dense (Dense)	(None, 200)	115400
dense_1 (Dense)	(None, 100)	20100
dense_2 (Dense)	(None, 1)	101

Total params: 144, 529
Trainable params: 144, 529
Non-trainable params: 0

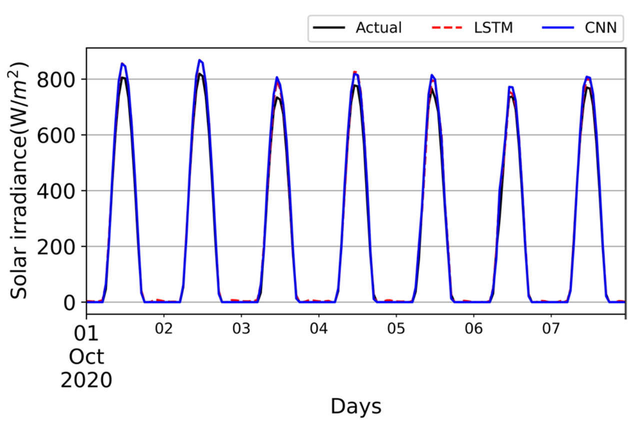


Fig. 20 One week GHI forecasting using seasonal model

Table 13 CNN-based model summary

Model: "sequentiL"

Layer (type)	Output Shape	Param #
conv1d (Conv1d)	(None, 70, 32)	128
max_pooling1d (Max_ Pooling1d)	(None, 70, 32)	0
flatten (Flatten)	(None, 2240)	0
dense (Dense)	(None, 200)	448200
Dense_1 (Dense)	(None, 100)	20100
Dense_2 (Dense)	(None, 1)	101

Total params: 468, 529
Trainable params: 468, 529
Non-trainable params: 0

took more time for training than the CNN-based model. This may be caused by the sequential nature of LSTM operations, where each time step depends on the previous one. This dependency limits the potential for parallelization, making LSTMs inherently slower to train than CNNs, which can effectively leverage parallel processing. Although the examined predicting schemes seem to exhibit significant training time, they can be easily implemented and instantly deployed in real-time GHI forecasting applications, resulting in efficient management of photovoltaic plants. Hence, according

Table 14 Training time and number of trainable parameters for the two best forecasting models

Model	Training time (hh:mm)	No. of trainable parameters
LSTM-based model	2:58	144529
CNN-based model	0:11	468529

to the prediction accuracy and computation complexity of each model, the most suitable model for the collected dataset could be selected.

Forecasting Uncertainty and Prediction Intervals

Solar irradiance forecasting is crucial, as inaccurate forecasts can significantly impact solar plant management. Forecast evaluation and accuracy metrics offer a method for generating confidence intervals (also known as prediction intervals) for forecasts. Including confidence intervals in forecasts provides an indicator of expected accuracy, replacing single deterministic prediction values with distributions or ranges of expected values. Additionally, prediction intervals are valuable as they provide a range within which the forecast is likely to fall, given a specified degree of confidence.

To estimate the prediction intervals, some assumptions about the distribution of the residuals are considered. For simplicity, we assume a normal distribution of forecast errors with a zero mean and a standard deviation equal to σ [45]. Confidence intervals are then expressed as multiples of the standard deviation corresponding to a given confidence level (e.g., 2σ for 95.4% confidence, 3σ for 99.7% confidence) [45]. In this study, a 95% prediction interval is used, where 95% of the data points fall within 1.96 standard deviations of the mean. Thus, we multiply 1.96 by the standard deviation σ to calculate the prediction interval size [46], as illustrated in Fig. 21. Figure 21 shows time-series plots of measured and modeled GHI with prediction bands for one-day forecasts. It can be easily observed that on clear days, the prediction bands are relatively narrow, indicating that GHI is highly predictable in clear-sky conditions. However, on days with more cloud cover, the irradiance becomes much less predictable, as shown by the wider prediction bands.

Conclusion

This paper investigates the prediction of global horizontal irradiance one hour ahead using two deep learning networks: CNN and LSTM. The study utilizes meteorological data from the National Renewable Energy Laboratory (NREL), focusing on Los Angeles for its diverse meteorological characteristics. Two prediction approaches are implemented: one based solely on historical GHI records and another that integrates historical GHI data with additional meteorological parameters. The research conducts various experiments to analyze both annual and seasonal models, focusing on factors such as the effect of timestep, the number of LSTM

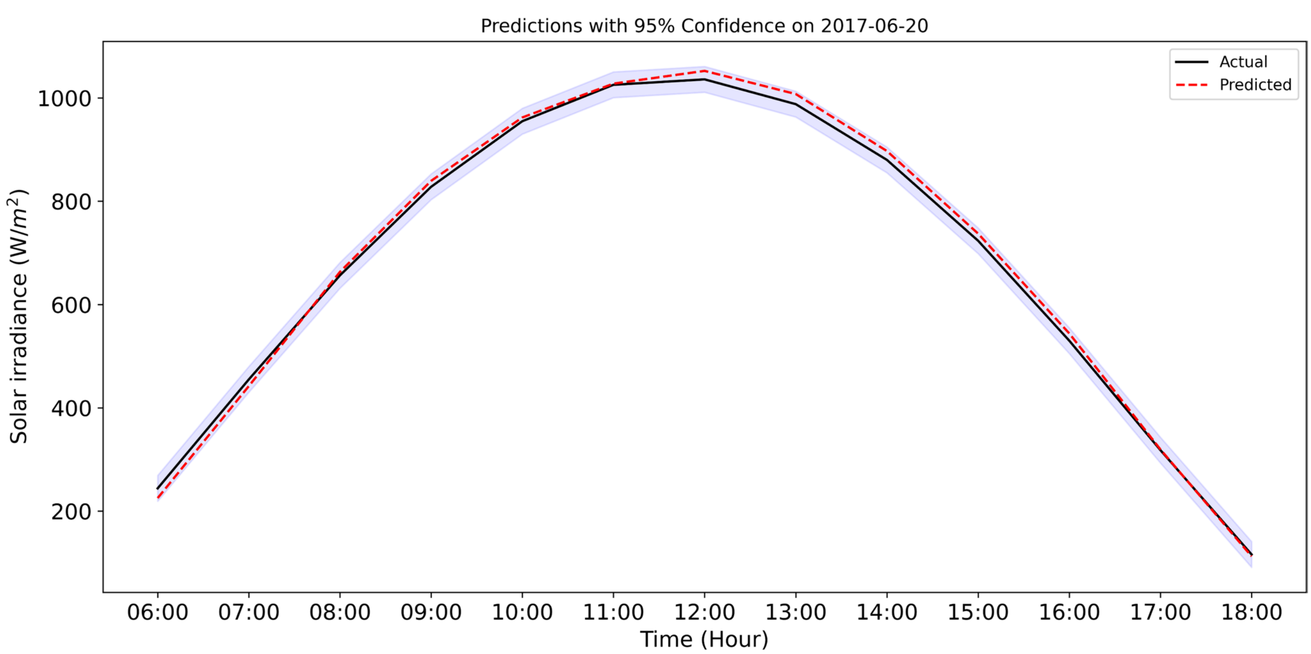


Fig. 21 One day hourly GHI Forecasts using LSTM-based model with prediction intervals for (a) clear sky, (b) partly cloudy and (c) cloudy days. The dark-bold lines are the measured values, the dashed red lines are the predictions, and the shaded regions

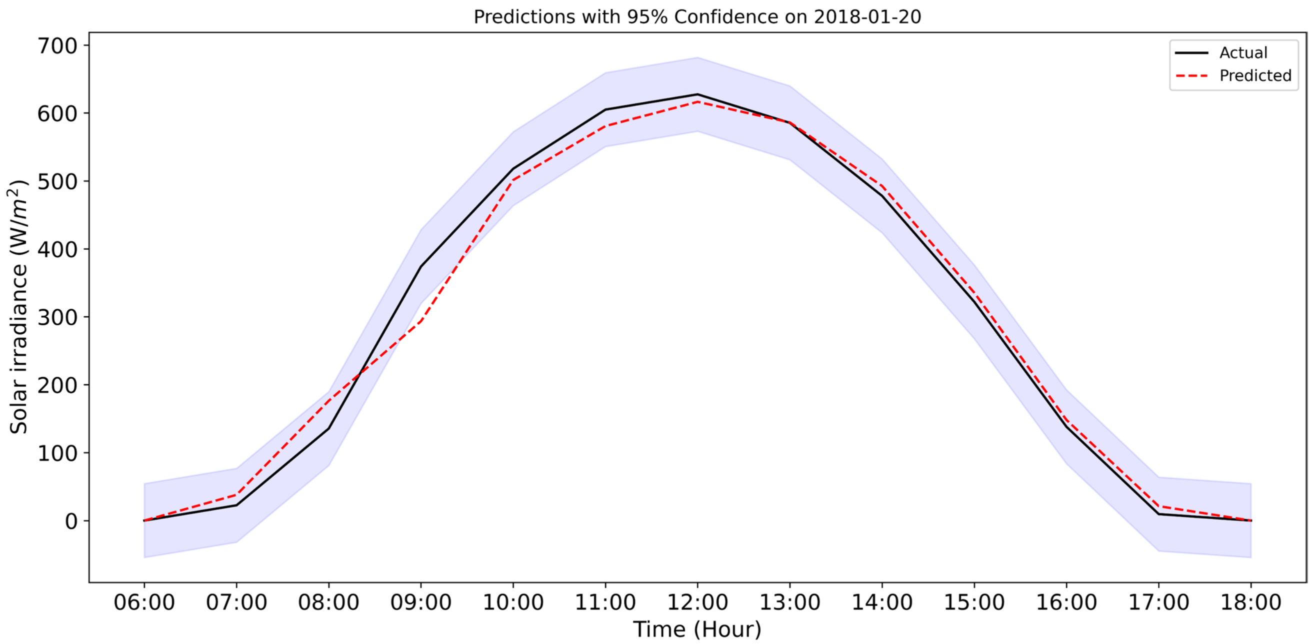


Fig. 21 (continued)

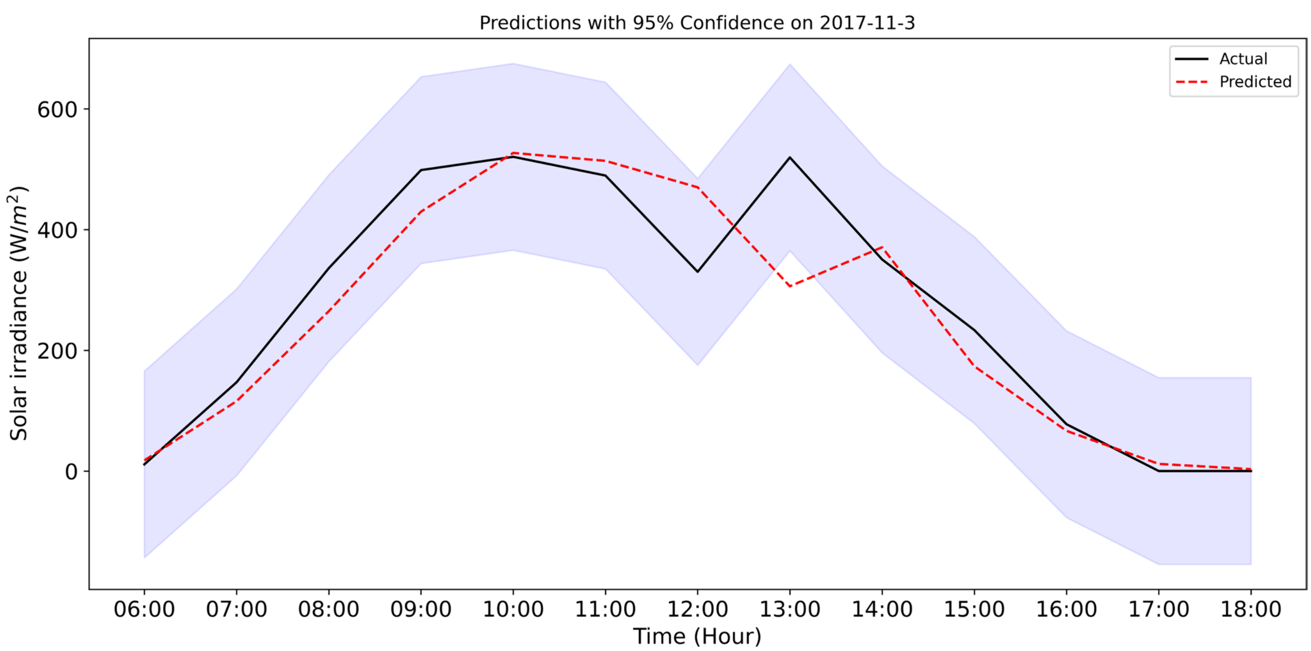


Fig. 21 (continued)

units, the impact of training dataset size, and the influence of input parameters and stacked layers for both LSTM and CNN.

Key findings reveal that utilizing a longer sliding window (N_step = 72 hours GHI) significantly enhances predictive capability; however, the number of units in the LSTM does not notably impact prediction accuracy. The quality of predictions is not solely determined by the length of the dataset. While longer time series data can improve the learning process, they may also introduce fluctuations and contradictory examples that can compromise accuracy compared to shorter and more stable datasets. Stacking LSTM layers can enhance accuracy, especially when combined with diverse input parameters, while stacking CNN layers does not necessarily yield improvements. A 1D CNN with max pooling set to 1 contributes to greater stability during training and better prediction. The obtained results show that the best annual LSTM-based model consistently outperforms the best CNN-based structure, achieving a root mean square error of 37.08W/m^2, and a mean absolute error of 15.23 W/m^2. Employing seasonal models has also been shown to enhance accuracy. Notably, while LSTM networks are effective at capturing long-term dependencies in time series data, it has been shown that a 1D CNN without max pooling may outperform LSTM models in terms of efficiency. A 95% prediction confidence interval is constructed, providing a range within which the forecast is likely to fall with a specified degree of confidence. Therefore, solar irradiance is highly predictable on clear days, whereas on days with more cloud cover, it becomes much less predictable, thereby enhancing the efficiency of solar plant management.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests All authors Tahar Bouadjila, Khaled Khelil, Djamel Rahem, and Farid Berrezzek declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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